Photometric Test Report

Reference Standards

ANSI C82.77-10-2014
IES LM-79-2019

Prepared For

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Test Laboratory Address:
No.2, Chengwan Road, Suzhou Industrial Park, Suzhou 215122, China
Catalog Number
BPANEL-50S54-UNV-D-8SC61-22-G-WH-P

Project Number
4791498147.3
Report Number
4791498147.3-2a

Test Date
2024-10-30~2024-11-05
Issue Date
2024-11-21
Revision Date
N/A

Prepared By	Approved By
Becky Fan	Roger Xu
Becky Fan	Roger Xu

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Test List

Sample Received Date: 2024-10-16

Test Item	Sample ID	Model Number	Tests Conducted By
Integrating Sphere Test	7712166	BPANEL-50S54-UNV-D-8SC61-22-G-WH-P	Hom Dong
Goniophotometer Test	7712166	BPANEL-50S54-UNV-D-8SC61-22-G-WH-P	Hom Dong
THD and PF Test	7712166	BPANEL-50S54-UNV-D-8SC61-22-G-WH-P	Hom Dong

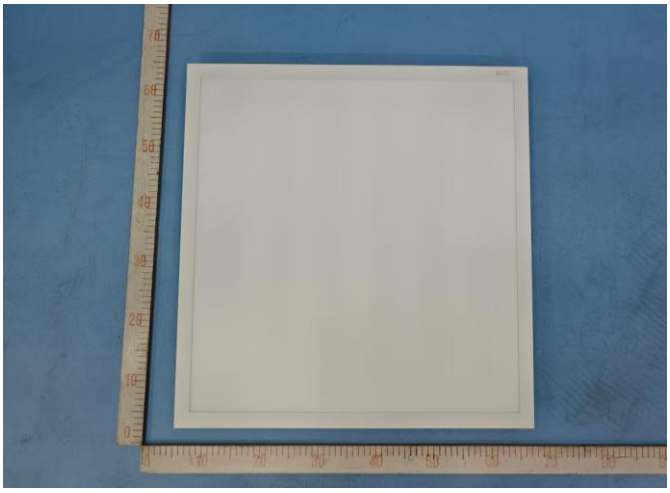
Remark (if any)

1. UL test equipment information is recorded on Meter Use in UL's Aurora database.

Product Description

Luminaire Description: 2x2 Luminaires for Ambient Lighting of Interior Commercial Spaces
Model Number: BPANEL-50S54-UNV-D-8SC61-22-G-WH-P
Electrical Ratings and CCT: 120-277Vac, 50/60 Hz, 50W max, 3000K/3500K/4000K/5000K/5700K/6500K tunable
Driver Model Number: BQ-50L6-1250-40-M56
LED Package: BXVN-30E-13H-9NV, BXVN-65E-13H-9NV, Bridgelux Inc
Family Model and Variation: BPANEL-50S54-UNV-D-8SC61-22-XX-YY-P-ZZZ-***-##, where "XX" represent mounting, can be G=Grid mounting, SF=Surface mounting, SP=Suspended mounting, DF=Drywall flange mounting. "YY" represent Finish color, can be WH=White or others. "ZZZ" represent Emergency, can be EM*= Emergency types(* can be blank or any letter), blank=No emergency. "***" represent Sensor receptacle, can be SR#= Sensor receptacle types(# can be blank or any number), blank= Without sensor receptacle. ## represent junction box, can be JB=with junction box, blank=without junction box.

Photos of Products Characteristics



Integrating Sphere Test

Model No.	BPANEL-50S54-UNV-D-8SC61-22-G-WH-P	Sample ID.	7712166
Operate time (Min.)	55	Stabilization time (Min.)	50

Test Method

1.The sample was tested according to the IES LM-79-2019, and the product is assume to be brand new without seasoning.
2.Photometric paramters were measured using an integrating sphere, a spectroradiometer and software. The ambient temperature condition inside the sphere was maintained at 25 °C ± 0.9 °C.The reference standard lamp is power 100W omni-directional Incandescent lamp and was calibrated by National Institute of Metrology, China.
3.The sample measurements were made using a spectroradiometer connected by a fiber optic cable and detector through the detector port of the integrating sphere. Coating reflectance of the integrating sphere was 90% to 98%. Photometric measurement conditions was using 4π geometry. The self-absorption factor is applied in the final test result. The sample was operated at rated voltage and was stabilized before measurement. Chromaticity coordinates, correlated color temperature and color rendering index were calculated from the spectral radiant flux measurements taken at 1 nm intervals over the range of 380 to 780 nm.

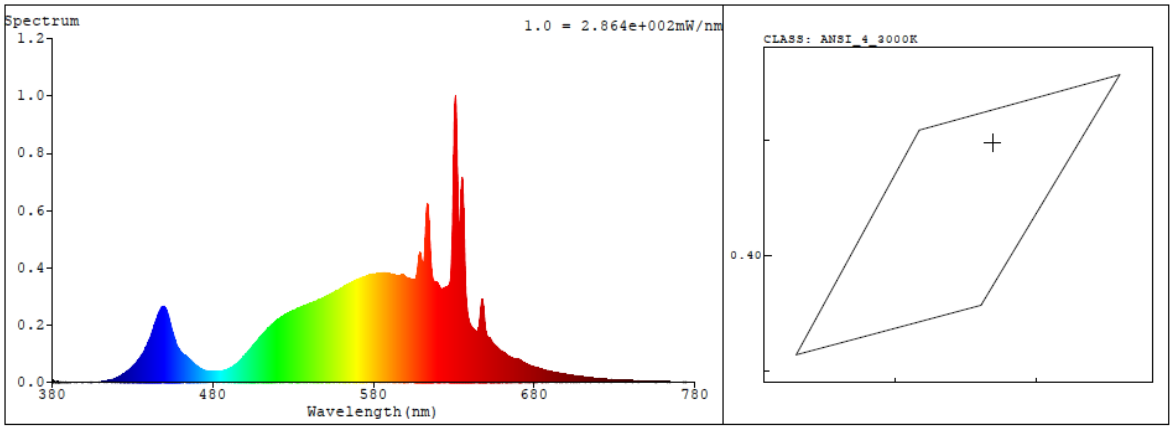
Integrating Sphere Test Conditions

Temperature (°C)	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	Orientation
25.1	120	60	0.400	47.59	0.9904	Horizontal

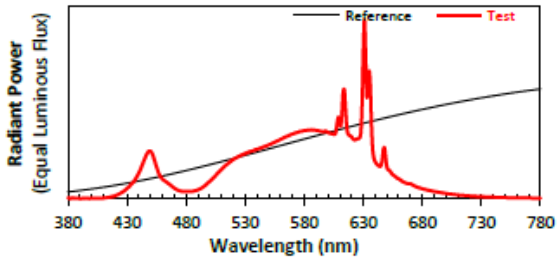
Test Results

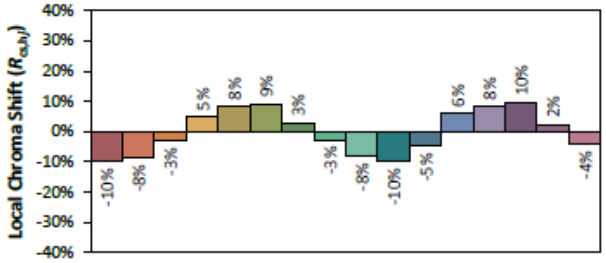
CCT (K)	CRI (Ra)	R9	Rf	Rg	Flux (lm)	Luminous Efficacy (lm/W)
3046	81	20	81	99	6286.3	132.1

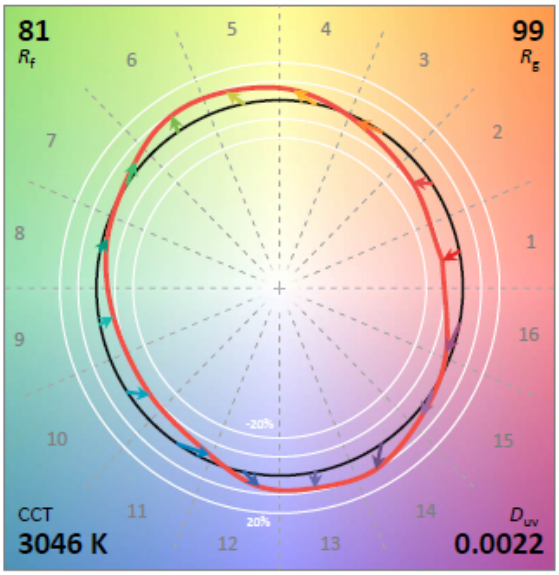
Spectrum

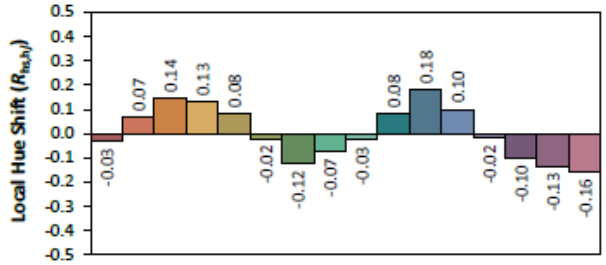


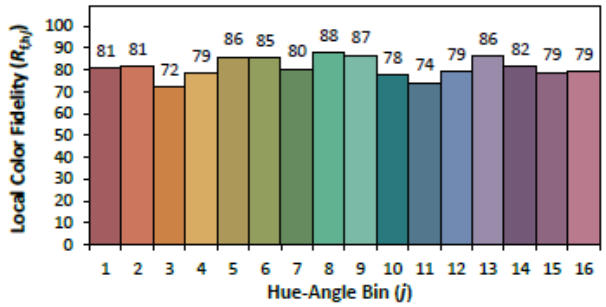
ANSI/IES TM-30-18 Color Rendition Report

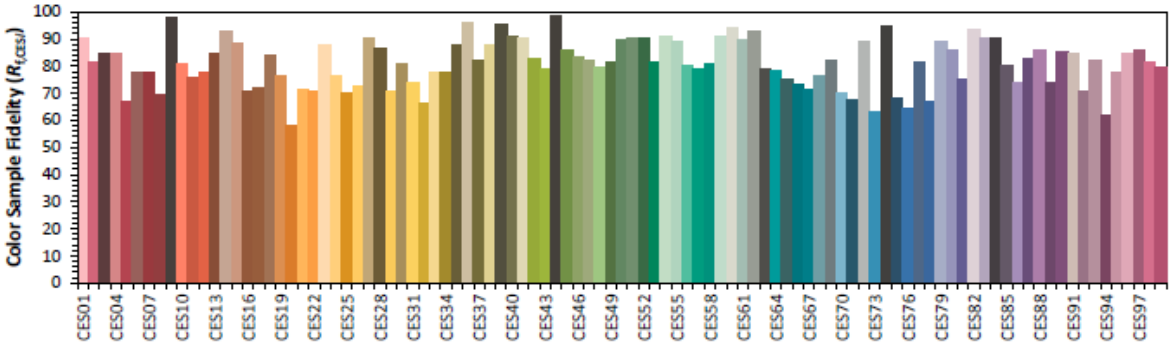












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

x	0.4369
y	0.4097
u'	0.2482
v'	0.5236

CIE 13.3-1995 (CRI)	
R_a	81
R_9	20

Colors are for visual orientation purposes only. Created with the ANSI/IES TM-30-18 Calculator Version 2.00.

Integrating Sphere Test

Model No.	BPANEL-50S54-UNV-D-8SC61-22-G-WH-P		Sample ID.	7712166
Operate time (Min.)	55		Stabilization time (Min.)	50

Test Method

1.The sample was tested according to the IES LM-79-2019, and the product is assume to be brand new without seasoning.
2.Photometric paramters were measured using an integrating sphere, a spectroradiometer and software. The ambient temperature condition inside the sphere was maintained at 25 °C ± 0.9 °C.The reference standard lamp is power 100W omni-directional Incandescent lamp and was calibrated by National Institute of Metrology, China.
3.The sample measurements were made using a spectroradiometer connected by a fiber optic cable and detector through the detector port of the integrating sphere. Coating reflectance of the integrating sphere was 90% to 98%. Photometric measurement conditions was using 4π geometry. The self-absorption factor is applied in the final test result. The sample was operated at rated voltage and was stabilized before measurement. Chromaticity coordinates, correlated color temperature and color rendering index were calculated from the spectral radiant flux measurements taken at 1 nm intervals over the range of 380 to 780 nm.

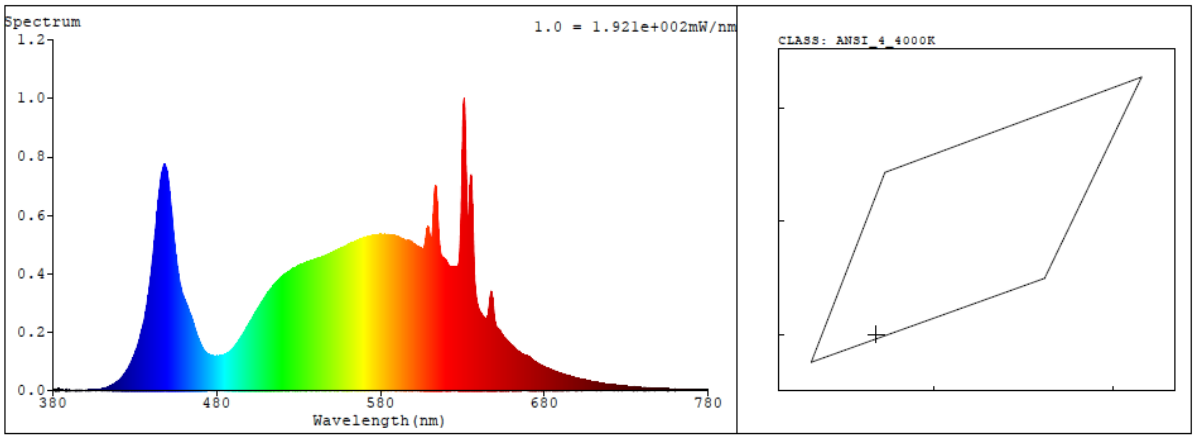
Integrating Sphere Test Conditions

Temperature (°C)	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	Orientation
25.1	120	60	0.399	47.48	0.9905	Horizontal

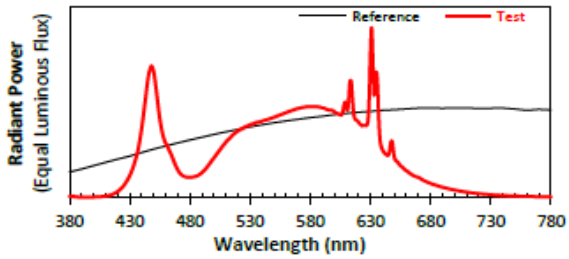
Test Results

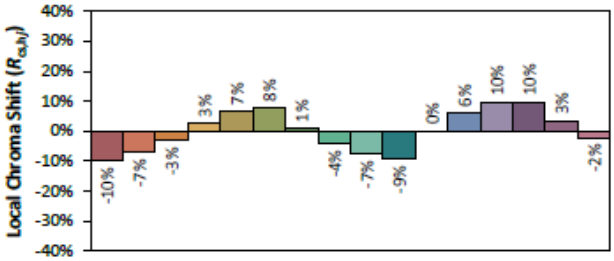
CCT (K)	CRI (Ra)	R9	Rf	Rg	Flux (lm)	Luminous Efficacy (lm/W)
4053	83	27	82	100	6384.4	134.5

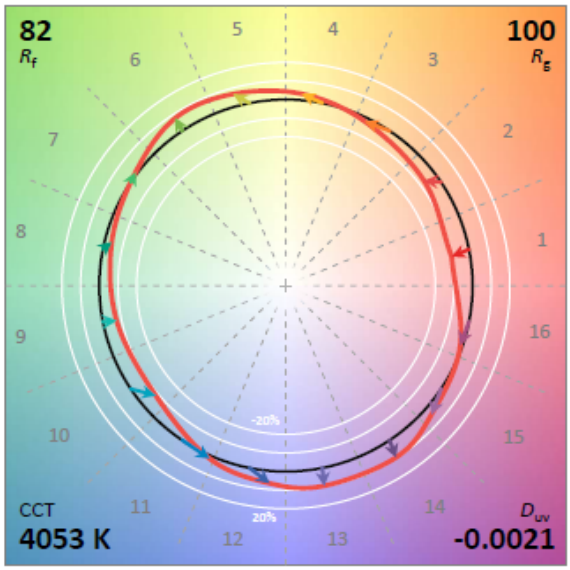
Spectrum

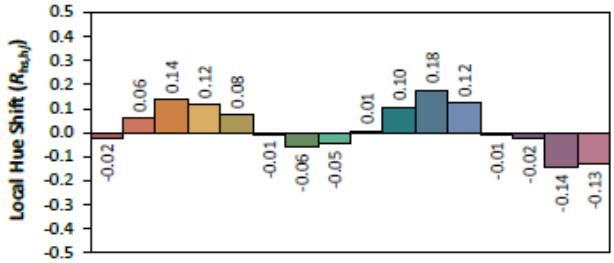


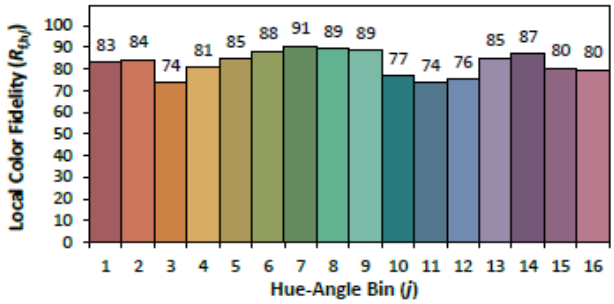
ANSI/IES TM-30-18 Color Rendition Report

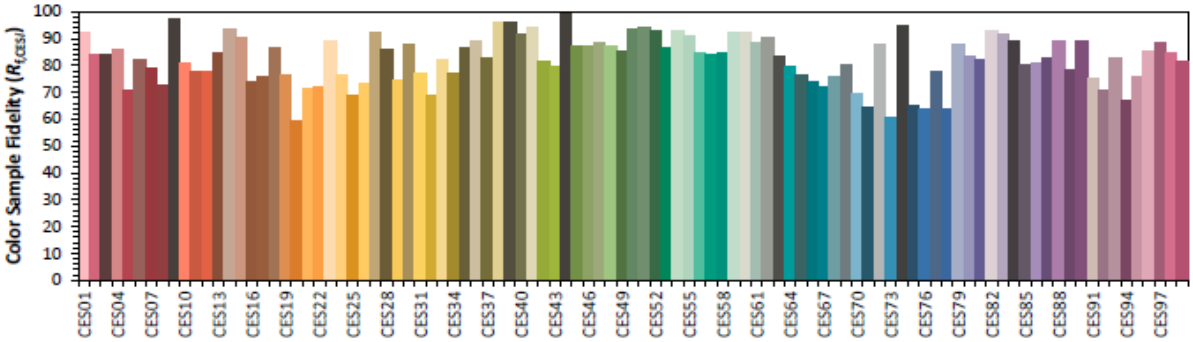












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

x	0.3767
y	0.3699
u'	0.2254
v'	0.4980

CIE 13.3-1995 (CRI)	
R_a	83
R_9	27

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Integrating Sphere Test

Model No.	BPANEL-50S54-UNV-D-8SC61-22-G-WH-P		Sample ID.	7712166
Operate time (Min.)	55		Stabilization time (Min.)	50

Test Method

1.The sample was tested according to the IES LM-79-2019, and the product is assume to be brand new without seasoning.
2.Photometric paramters were measured using an integrating sphere, a spectroradiometer and software. The ambient temperature condition inside the sphere was maintained at 25 °C ± 0.9 °C.The reference standard lamp is power 100W omni-directional Incandescent lamp and was calibrated by National Institute of Metrology, China.
3.The sample measurements were made using a spectroradiometer connected by a fiber optic cable and detector through the detector port of the integrating sphere. Coating reflectance of the integrating sphere was 90% to 98%. Photometric measurement conditions was using 4π geometry. The self-absorption factor is applied in the final test result. The sample was operated at rated voltage and was stabilized before measurement. Chromaticity coordinates, correlated color temperature and color rendering index were calculated from the spectral radiant flux measurements taken at 1 nm intervals over the range of 380 to 780 nm.

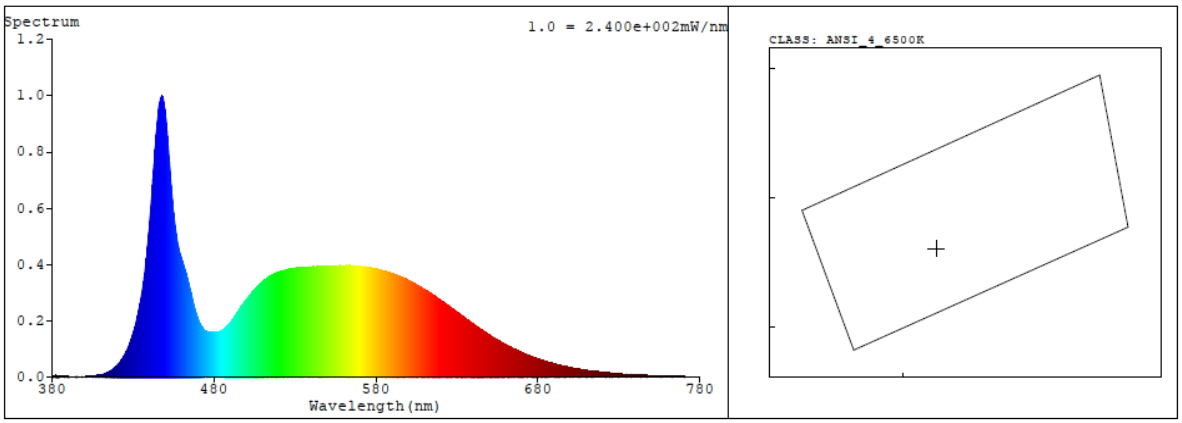
Integrating Sphere Test Conditions

Temperature (°C)	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	Orientation
25.1	120	60	0.402	47.74	0.9903	Horizontal

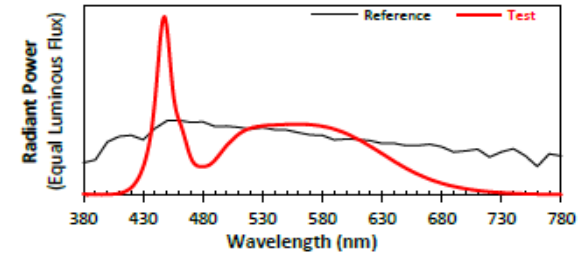
Test Results

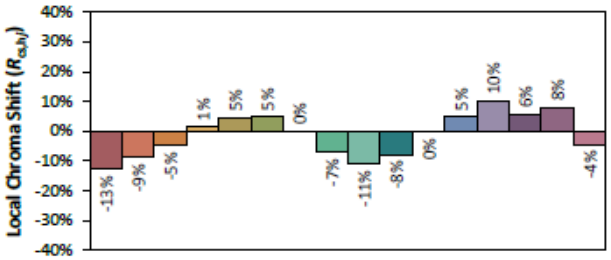
CCT (K)	CRI (Ra)	R9	Rf	Rg	Flux (lm)	Luminous Efficacy (lm/W)
6612	81	5	81	97	6340.8	132.8

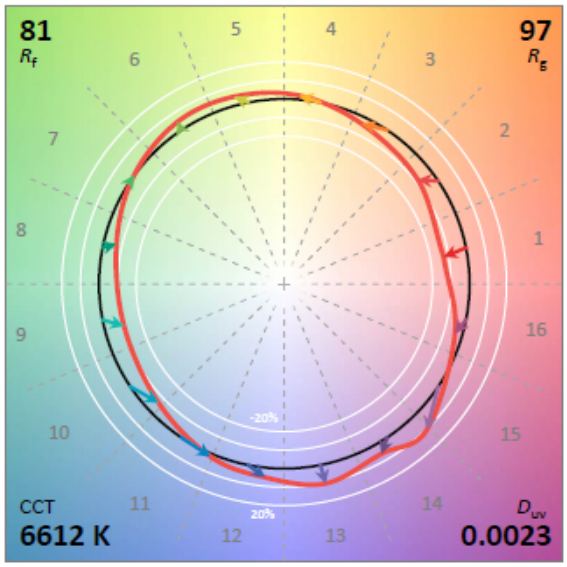
Spectrum

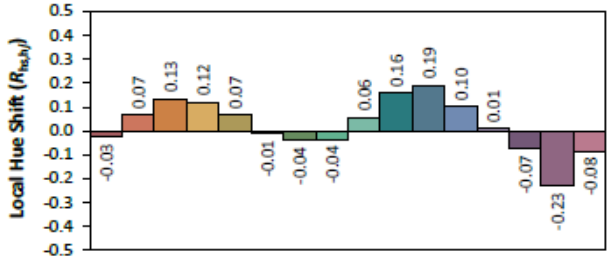


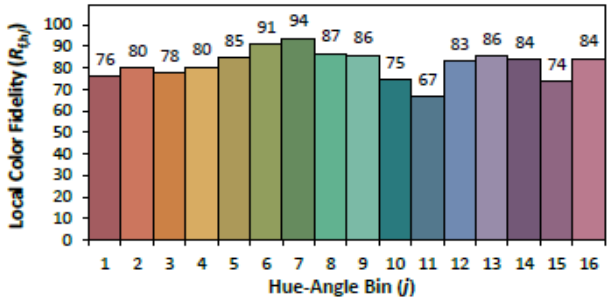
ANSI/IES TM-30-18 Color Rendition Report

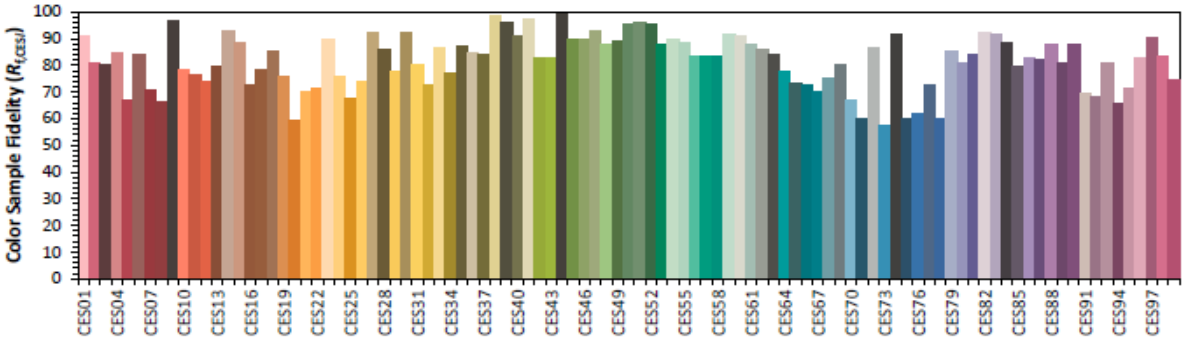












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

x	0.3112
y	0.3258
u'	0.1980
v'	0.4664

CIE 13.3-1995 (CRI)	
R_a	81
R_9	5

Colors are for visual orientation purposes only. Created with the ANSI/IES TM-30-18 Calculator Version 2.00.

Goniophotometer Test

Model No.	BPANEL-50S54-UNV-D-8SC61-22-G-WH-P	Sample ID.	7712166
Operate time (Min.)	65	Stabilization time (Min.)	50

Test Method

1.The sample was tested according to the IES LM-79-2019, and the product is assume to be brand new without seasoning.
2.Photometric parameters were measured using a type C goniophotometer and software.
3.The ambient temperature shall be maintained at 25 °C ± 0.9 °C, measured at a point not more than 1.5 m from the sample and at the same height as the sample. The reference standard lamp is power 400W omni-directional Incandescent lamp and was calibrated by National Institute of Metrology, China.
4.The samples were operated at rated voltage and was stabilized before measurement. Luminous flux, luminaire efficacy, zonallumen were calculated from the software taken at 1° vertical intervals and 22.5° horizontal intervals. Photometric distance was more than five times of the largest dimension of the test SSL product.

Goniophotometer Test Conditions

Temperature (°C)	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	Orientation
24.3	120	60	0.401	47.72	0.9919	Horizontal

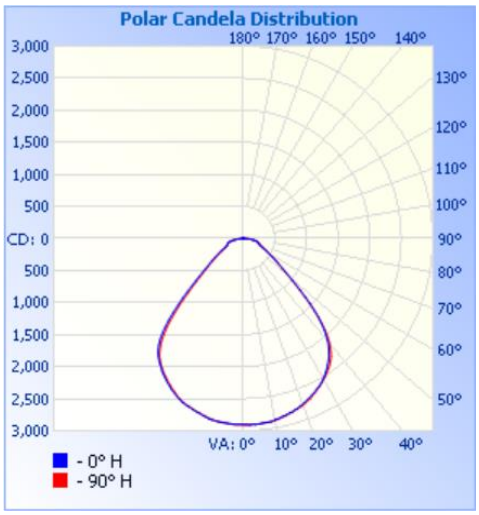
Test Results

Zonal Lumen Requirement (0°-60°)	Beam Angle (50%)		Spacing Criteria (0-180°)	Spacing Criteria (90°-270°)	Flux (lm)	Luminous Efficacy (lm/W)
	Horizontal Spread	Vertical Spread				
86.5%	90.1	91.2	1.30	1.30	6336.4	132.8

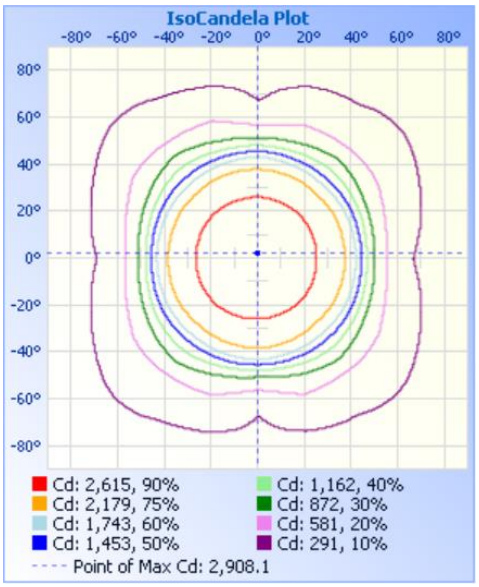
UGR
20.3

Goniophotometer Test (Cont'd)

Polar Candela Distribution



IsoCandela Plot



Goniophotometer Test (Cont'd)

Zonal Lumen Summary

Zonal Lumen Summary		
Zone	Lumens	% Luminaire
0-30	2,285.3	36.1%
0-40	3,719.7	58.7%
0-60	5,480.0	86.5%
60-90	841.0	13.3%
70-100	417.1	6.6%
90-120	5.4	0.1%
0-90	6,321.0	99.8%
90-180	15.4	0.2%
0-180	6,336.4	100%

Lumens Per Zone

Lumens Per Zone					
Zone	Lumens	% Total	Zone	Lumens	% Total
0-10	275.6	4.3%	90-100	2.8	0%
10-20	794.4	12.5%	100-110	1.2	0%
20-30	1,215.3	19.2%	110-120	1.3	0%
30-40	1,434.4	22.6%	120-130	1.6	0%
40-50	1,115.6	17.6%	130-140	2.0	0%
50-60	644.7	10.2%	140-150	2.2	0%
60-70	426.7	6.7%	150-160	2.0	0%
70-80	291.2	4.6%	160-170	1.5	0%
80-90	123.1	1.9%	170-180	0.6	0%

Goniophotometer Test (Cont'd)

Intensity Data(cd)

	0	22.5	45	67.5	90	112.5	135	157.5	180	202.5	225	247.5	270	292.5	315	337.5	360
0	2903	2903	2903	2903	2903	2903	2903	2903	2903	2903	2903	2903	2903	2903	2903	2903	2903
1	2899	2906	2903	2899	2905	2903	2903	2903	2899	2902	2898	2899	2899	2898	2899	2902	2904
2	2908	2898	2905	2903	2901	2902	2902	2900	2905	2897	2898	2900	2900	2905	2901	2900	2899
3	2904	2897	2903	2906	2899	2903	2900	2902	2899	2894	2895	2896	2904	2903	2896	2902	2904
4	2904	2896	2899	2899	2904	2896	2899	2894	2896	2891	2894	2897	2902	2897	2895	2900	2902
5	2900	2899	2898	2898	2902	2898	2895	2891	2893	2893	2888	2889	2894	2891	2889	2895	2899
6	2897	2891	2897	2898	2894	2898	2891	2886	2890	2886	2891	2887	2885	2884	2886	2892	2894
7	2888	2890	2894	2892	2893	2889	2887	2892	2888	2885	2880	2884	2884	2887	2885	2886	2889
8	2891	2891	2884	2888	2892	2892	2882	2879	2880	2879	2875	2879	2881	2884	2881	2884	2883
9	2886	2885	2883	2886	2890	2886	2881	2879	2879	2871	2870	2874	2871	2874	2870	2880	2882
10	2870	2874	2880	2878	2876	2873	2872	2874	2868	2866	2866	2863	2867	2864	2872	2869	2877
11	2865	2867	2865	2866	2870	2868	2861	2853	2860	2852	2851	2851	2857	2856	2853	2863	2867
12	2848	2857	2858	2862	2861	2849	2851	2847	2847	2841	2842	2840	2840	2839	2844	2847	2849
13	2845	2840	2844	2843	2843	2846	2835	2833	2832	2826	2822	2820	2831	2830	2833	2829	2842
14	2830	2834	2834	2835	2839	2828	2823	2819	2820	2815	2812	2812	2809	2811	2813	2824	2826
15	2818	2815	2819	2822	2820	2820	2812	2810	2803	2799	2801	2794	2804	2803	2806	2806	2818
16	2811	2809	2811	2811	2816	2801	2799	2792	2794	2789	2779	2784	2787	2786	2789	2801	2798
17	2797	2789	2798	2801	2795	2798	2787	2783	2778	2772	2771	2767	2774	2773	2780	2779	2792
18	2784	2783	2785	2793	2790	2777	2777	2767	2762	2762	2753	2758	2759	2762	2762	2774	2777
19	2764	2766	2772	2773	2776	2774	2761	2760	2755	2744	2742	2741	2750	2745	2755	2753	2762
20	2747	2757	2765	2767	2766	2751	2753	2738	2735	2734	2727	2729	2732	2736	2739	2747	2753
25	2647	2656	2664	2663	2665	2653	2646	2637	2630	2626	2618	2618	2613	2619	2628	2638	2649
30	2527	2522	2523	2538	2536	2526	2514	2500	2491	2474	2462	2467	2472	2470	2476	2496	2508
35	2340	2351	2348	2380	2387	2355	2327	2327	2326	2293	2244	2271	2301	2285	2280	2315	2352
40	2071	2036	1989	2046	2116	2048	1952	1998	2047	1928	1838	1900	1960	1903	1839	1959	2073
45	1593	1501	1474	1526	1586	1515	1436	1453	1459	1388	1348	1354	1345	1340	1334	1403	1499
50	1007	1030	1119	1070	996	1062	1091	997	900	966	1020	909	837	889	1005	971	926
55	670	738	831	756	668	740	811	714	612	706	738	658	570	649	725	700	619
60	475	599	576	601	482	589	563	585	435	579	499	547	409	541	498	566	437
65	336	503	412	516	355	506	406	495	318	486	378	472	304	467	379	482	319
70	265	409	349	429	285	424	350	400	267	388	320	377	264	376	318	390	265
75	237	285	283	309	249	312	284	291	243	281	275	271	239	268	271	285	238
80	187	196	232	213	197	217	238	201	181	197	227	194	173	190	224	198	180
85	108	117	151	133	117	138	153	117	94	105	123	102	85	102	122	109	94
90	21	22	26	32	31	32	23	13	5	5	7	6	5	7	7	6	7
95	0	1	1	2	2	1	1	1	1	2	2	1	1	1	1	1	1
100	1	1	1	1	2	1	1	1	1	1	1	1	1	1	1	1	1
105	2	1	1	1	1	1	1	1	1	1	1	1	2	2	1	1	1
110	1	2	1	1	1	1	2	2	1	1	1	2	1	1	1	1	1
115	1	1	1	1	1	1	2	2	1	2	2	1	1	1	1	1	1
120	2	1	2	2	1	1	1	2	1	2	2	2	2	2	2	2	2
125	2	2	2	2	2	2	2	2	2	1	2	2	2	2	2	2	2
130	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
135	2	2	3	2	3	2	3	2	3	3	3	3	3	2	2	3	2
140	3	3	3	3	3	3	3	3	3	4	3	3	3	3	3	3	3
145	4	4	4	4	3	3	3	4	3	4	3	4	4	4	4	3	4
150	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4
155	4	4	5	5	4	4	4	5	4	5	4	4	4	5	4	4	5
160	4	5	5	5	4	4	5	4	4	5	5	5	5	5	5	5	5
165	5	6	6	6	6	6	5	5	5	6	6	6	6	6	6	5	6
170	6	6	6	6	6	6	6	6	6	7	7	7	6	7	6	7	6
175	6	7	7	7	6	7	7	7	7	7	7	7	7	7	6	7	7
180	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7

Corrected UGR values generated per CIE 190-2010

UGR Table - Corrected

Reflectances											
Ceiling Cavity		70	70	50	50	30	70	70	50	50	30
Walls		50	30	50	30	30	50	30	50	30	30
Floor Cavity		20	20	20	20	20	20	20	20	20	20
Room Size		UGR Viewed Crosswise					UGR Viewed Endwise				
X=2H	Y=2H	15.2	16.7	15.6	17.0	17.3	15.4	16.9	15.8	17.2	17.5
	3H	16.8	18.1	17.2	18.4	18.8	17.1	18.4	17.5	18.7	19.1
	4H	17.5	18.7	17.9	19.1	19.4	17.8	19.0	18.2	19.4	19.8
	6H	18.2	19.3	18.6	19.7	20.1	18.5	19.7	19.0	20.0	20.4
	8H	18.5	19.6	18.9	20.0	20.4	18.9	20.0	19.3	20.3	20.7
	12H	18.7	19.8	19.2	20.1	20.6	19.2	20.2	19.6	20.6	21.0
4H	2H	15.7	16.9	16.1	17.2	17.6	15.9	17.1	16.3	17.4	17.8
	3H	17.6	18.7	18.0	19.1	19.5	18.0	19.0	18.4	19.4	19.8
	4H	18.6	19.5	19.0	19.9	20.3	18.9	19.9	19.4	20.3	20.7
	6H	19.4	20.3	19.9	20.7	21.2	19.8	20.7	20.3	21.1	21.6
	8H	19.8	20.6	20.3	21.0	21.5	20.3	21.0	20.7	21.5	21.9
	12H	20.1	20.8	20.6	21.3	21.8	20.6	21.3	21.1	21.8	22.2
8H	4H	19.0	19.8	19.5	20.2	20.7	19.4	20.1	19.8	20.6	21.0
	6H	20.1	20.7	20.6	21.2	21.7	20.5	21.1	21.0	21.6	22.1
	8H	20.6	21.2	21.1	21.7	22.2	21.0	21.6	21.5	22.1	22.6
	12H	21.0	21.5	21.5	22.0	22.6	21.5	22.1	22.1	22.6	23.1
12H	4H	19.1	19.8	19.6	20.2	20.7	19.4	20.1	19.9	20.6	21.0
	6H	20.3	20.8	20.8	21.3	21.8	20.6	21.2	21.2	21.7	22.2
	8H	20.8	21.4	21.4	21.8	22.4	21.3	21.8	21.8	22.3	22.8

Maximum UGR = 23.1

THD and PF Test

Model No.	BPANEL-50S54-UNV-D-8SC61-22-G-WH-P	Sample ID.	7712166

Test Method

1. The samples were tested according to the ANSI C82.77-10-2014.
2. The ambient temperature condition was maintained at 25 °C ± 1 °C. The sample measurement was made using a digital power meter and power supply. The sample was operated at rated voltage and stabilized before measurement. The total harmonic distortion were calculated from the digital power meter.

Test Results

CCT Range	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	Current THD	Orientation
0%-3000K	120	60	0.401	47.7	0.9919	11.60%	Horizontal
	277	60	0.176	46.7	0.9585	10.12%	Horizontal
20%-3500K	120	60	0.400	47.6	0.9918	11.06%	Horizontal
	277	60	0.175	46.6	0.9585	10.01%	Horizontal
40%-4000K	120	60	0.400	47.7	0.9918	11.00%	Horizontal
	277	60	0.175	46.6	0.9586	9.95%	Horizontal
60%-5000K	120	60	0.400	47.7	0.9918	11.37%	Horizontal
	277	60	0.176	46.6	0.9587	9.97%	Horizontal
80%-5700K	120	60	0.401	47.7	0.9918	11.03%	Horizontal
	277	60	0.176	46.6	0.9585	9.98%	Horizontal
100%-6500K	120	60	0.402	47.9	0.9918	10.96%	Horizontal
	277	60	0.176	46.8	0.9591	9.87%	Horizontal

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